

Fall 2025

QFGB8914: Derivatives

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Synopsis

This course provides a comprehensive introduction to financial derivatives, covering their structure, pricing, and applications across asset classes. Beginning with foundational instruments such as forwards, futures, swaps, and options, students explore the economic functions of derivatives in hedging, speculation, and risk transfer. The course builds technical proficiency through the application of no-arbitrage pricing, term structure models, and real-world market conventions.

Midway, the curriculum advances into interest rate and fixed income derivatives, including SOFR futures, bond futures, forward rate agreements, and interest rate swaps. Students learn to construct hedges, calculate duration and DV01, and manage exposure across different maturities and credit qualities.

The second half of the course transitions into option theory, with in-depth exploration of European and American options, the binomial model, and risk-neutral valuation. A rigorous derivation of the Black-Scholes equation introduces continuous-time pricing, which is then applied to volatility products, VIX derivatives, and variance swaps.

Advanced modules address implied volatility surfaces, stochastic volatility models (e.g., SVI), the Bachelier model, and applications in commodities and spreads.

Topics

Week	Topic
1	Introduction
2	Forwards and Futures
3	No-arbitrage pricing
4	Equity futures
5	Interest rate futures

6	Forward rate agreements and interest rate swaps
7	MidTerm
8	European Options
9	American Options
10	Ito's Lemma and Black Scholes Equation
11	Variances Swap and Correlation Swap
12	Options Markets
13	Implied Volatility Surface
14	Bachelier Model
15	Final Exam

Optional Textbook

Options, Futures, and Other Derivatives by John Hull

Quizzes and Exams

1. Short quizzes given periodically through-out the term to test your knowledge acquisition.
2. Midterm will be given on Oct. 14th
3. Final exam will be given on Dec. 16th

Assignments

There will be homework assignments throughout the course. Students are encouraged to work in works to enhance learning, open discussion, as well as social interactions. Students are required to submit each assignment on time. Late submission will not be accepted.

Class Participation

Attendance will be taken randomly during the semester. Students are strongly encouraged to actively ask questions, give feedback to the professor, and participate in class discussion. Both attendance and participation will count toward final grade.

Grading Policy

- Quizzes and homework assignments: 25%
- Midterm: 25%
- Final Exam: 35%
- Participation: 15%
 - o Attendance 5%
 - o Active participation 10%

Academic Integrity

Students are expected to abide by the Honor Code of Fordham University Graduate School of Business and maintain the highest standards with regard to honesty, effort, and performance. I take the matters of academic integrity very seriously and expect that you do, too. Any attempt to represent the work of others (including copying solutions to similar problems from previous semesters or online) as your own will be considered plagiarism. Discipline will be enforced rigorously according to the policies of Fordham University.